Fractional indices

1 Write the following in surd form:

a $x^{\frac{1}{3}}$ **b** $a^{\frac{3}{2}}$ **c** $20^{\frac{5}{4}}$ **d** $y^{-\frac{1}{2}}$

e $3x^{\frac{1}{2}}$ **f** $b^{2\frac{1}{3}}$ **g** $y^{1.5}$ **h** $5^{-\frac{4}{3}}$

2 Write the following as a single power of 2:

a $\sqrt[3]{2}$ **b** $\sqrt[5]{16}$ **c** $\sqrt{32}$ **d** $2\sqrt{2}$

e $8\sqrt[3]{2}$ **f** $2^a \times \sqrt{2}$ **g** $\dfrac{8}{\sqrt{2}}$ **h** $\dfrac{1}{\sqrt{8}}$

3 Write the following as a single power of 3:

a $\sqrt{27}$ **b** $3^x \times \sqrt[4]{3}$ **c** $9\sqrt{3}$ **d** $\dfrac{1}{\sqrt[3]{9^x}}$

4 Complete the following statement:

$$\begin{aligned}\sqrt{m^8} &= \left(m^8\right)^{\square} \\ &= m^{\square \times \frac{1}{2}} \\ &= m^{\square}\end{aligned}$$

5 Write the following in the form x^k, where k is rational:

a $\sqrt{x}$ **b** $\sqrt[6]{x}$ **c** $\dfrac{1}{\sqrt{x}}$ **d** $\left(\sqrt[3]{x}\right)^2$

e $\sqrt[4]{x^3}$ **f** $\left(\sqrt[7]{x}\right)^6$ **g** $\sqrt[4]{x^8}$ **h** $\dfrac{1}{\sqrt{x}}$

i $\sqrt{\sqrt{x}}$ **j** $\sqrt[3]{x^{6a}}$ **k** $\dfrac{1}{\sqrt{x^{-3}}}$ **l** $x\sqrt{x}$

m $\dfrac{1}{\sqrt[4]{x^5}}$ **n** $\sqrt[5]{\sqrt{x}}$ **o** $x^2 \times \sqrt[3]{x}$

6 Consider the expression $64^{\frac{2}{3}}$.

a Complete the statement: $64^{\frac{2}{3}} = \left(\sqrt[\square]{64}\right)^{\square}$ **b** Hence, evaluate $64^{\frac{2}{3}}$.

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7 Without using a calculator, evaluate the following:

a $1^{\frac{1}{10}}$ **b** $4^{\frac{3}{2}}$ **c** $121^{\frac{1}{2}}$ **d** $\sqrt[3]{27}$

e $125^{\frac{1}{3}}$ **f** $(-64)^{\frac{1}{3}}$ **g** $64^{-\frac{1}{6}}$ **h** $81^{-\frac{3}{4}}$

i $\sqrt[3]{-64}$ **j** $\left(\frac{9}{100}\right)^{\frac{1}{2}}$ **k** $\left(64^{\frac{1}{9}}\right)^{\frac{9}{2}}$ **l** $\left(\frac{8}{125}\right)^{\frac{2}{3}}$

m $(-32)^{\frac{4}{5}}$ **n** $\frac{64^{\frac{1}{3}}}{64^{\frac{2}{3}}}$ **o** $\frac{1}{-\sqrt{169}}$ **p** $\sqrt[3]{\frac{-64}{125}}$

q $\frac{\sqrt[3]{40}}{\sqrt[3]{5}}$ **r** $36^{\frac{1}{2}} - 32^{\frac{3}{5}}$ **s** $1000^{\frac{8}{9}} \times 1000^{-\frac{5}{9}}$ **t** $\sqrt[3]{\sqrt[4]{16} + \sqrt{625}}$

8 Use a calculator to evaluate the following, to two decimal places:

a $10^{\frac{3}{2}}$ **b** $4^{\frac{5}{3}}$ **c** $\sqrt[4]{5^3}$ **d** $\left(\sqrt[3]{12}\right)^5$

9 Simplify the following expressions, giving your answers in index form. Assume that all variables represent positive numbers.

a $\left(\sqrt{b}\right)^8$ **b** $\sqrt{m^6}$ **c** $\sqrt[4]{a^5}$ **d** $\frac{1}{\sqrt[5]{a^6}}$

e $\sqrt[3]{x^6}$ **f** $\sqrt[3]{m^3}$ **g** $\sqrt{j^2x^4}$ **h** $\left(\sqrt[4]{x^5y^3}\right)^{24}$

i $\left(4a^8\right)^{\frac{1}{2}}$ **j** $\left(625u^{16}v^{12}\right)^{\frac{1}{2}}$ **k** $8b^{\frac{3}{4}} \div 2b^{\frac{2}{3}}$ **l** $y^3 \times \sqrt[3]{y}$

m $\sqrt{(2x+9)^2}$ **n** $\sqrt[3]{9^3x^{18}y^{12}}$ **o** $\left(\frac{x^{15}}{1024}\right)^{\frac{2}{5}}$ **p** $\sqrt{\frac{36x^{18}}{y^{20}}}$

q $\sqrt{16y^2 + 24y + 9}$ **r** $\sqrt{x^2y^2 + 18xy^2 + 81y^2}$

10 Simplify the following expressions, giving your answers in surd form. Assume that all variables represent positive numbers.

a $\frac{\sqrt{x^2+5x+6}}{\sqrt{x+2}}$ **b** $\frac{\sqrt[3]{x^2+9x+20}}{\sqrt[3]{x+4}}$

11 Describe how we could interpret the expression $m^{\frac{q}{r}}$ in terms of powers and roots of m.

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12 Determine whether the following statements accurately describe the meaning of the expression $x^{-\frac{y}{z}}$:

a $x^{-\frac{y}{z}}$ means we are raising x to the power of $\frac{z}{y}$, then taking the reciprocal of the result.

b $x^{-\frac{y}{z}}$ means we are taking the reciprocal of x, then raising the result to the power of $\frac{y}{z}$.

c $x^{-\frac{y}{z}}$ means we are taking the reciprocal of x, then raising the result to the power of $\frac{z}{y}$.

d $x^{-\frac{y}{z}}$ means we are raising x to the power of $\frac{y}{z}$, then taking the reciprocal of the result.

13 Is there a real number that equals $\sqrt[4]{-16}$? Explain your answer.

14 Consider the expression $m^5 \times m\sqrt{m}$.

a Express it in simplest index form.

b Express it in surd form.

15 Consider the expression $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$.

a Complete the following working:

$$m^{\frac{1}{4}} = (m^{\square})^{\frac{1}{2}}$$
$$= \sqrt{\square}^{\frac{1}{2}}$$

b Hence, or otherwise, solve the equation $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$.

16 Consider the equation $x^{\frac{2}{3}} = 9$.

a Complete the statement: $\left(x^{\frac{2}{3}}\right)^{\square} = x$

b Hence, solve the equation $x^{\frac{2}{3}} = 9$.

17 Find the missing values in the following equations:

a $16^{\frac{3}{\square}} = 64$

b $(\square)^{\frac{2}{3}} = 25$

c $625^{\frac{3}{\square}} = 125$

d $(\square)^{\frac{4}{3}} = 81$

18 Solve the following equations for x:

a $x^{\frac{3}{4}} = 125$

b $(x-4)^{0.5} = 4$

c $(2x-1)^{\frac{2}{3}} = 9$

d $(4x+3)^{-1.5} = 8$

19 Solve the following equation for k:

$$\sqrt[k]{y} \times \sqrt[k]{y} \times \sqrt[k]{y} = y^{\frac{1}{2}}$$

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20 To evaluate $81^{\frac{3}{2}}$ would it be more efficient to use the property $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$, or the property $a^{\frac{m}{n}} = \sqrt[n]{a^m}$. Explain your answer.

Applications

21 The surface area A of a cube with a volume V is given by $A = 6V^{\frac{2}{3}}$.

- **a** Find the surface area of a cube with a volume of $8\ \text{cm}^3$.
- **b** Find the volume of a cube with a surface area of $216\ \text{cm}^2$.

22 The volume V of a sphere with surface area A is given by:

$$V = \frac{A^{\frac{3}{2}}}{6\sqrt{\pi}}$$

- **a** Find the volume of a sphere with a surface area of $50\ \text{cm}^2$. Round you answer to the nearest cubic centimetre.
- **b** Find the surface area of a sphere with a volume of $125\ \text{cm}^3$. Round you answer to the nearest square centimetre.

23 The volume V of a regular tetrahedron with edge length a is given by:

$$V = \frac{a^3}{6\sqrt{2}}$$

- **a** Find the volume of a tetrahedron with a side length of $2\ \text{cm}$.
- **b** Rearrange the formula to make a the subject.
- **c** Hence, find the side length of a tetrahedon with a volume of $72\ \text{cm}^3$.

24 A company's net profit (in dollars) can be modelled by the equation:

$$P = 50n^{\frac{3}{4}} - 2500$$

Where n is the number of units sold.

- **a** Find its net profit be in dollars if it manages to sell 256 units.
- **b** Find the number of units needed to be sold to make a net profit of $\$8300$.

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25 A company's net profit (in dollars) can be modelled by the equation:

$$P = 180n^{\frac{3}{5}} - 4000$$

Where n is the number of units sold.

a Find its net profit be in dollars if it manages to sell 32 units.

b Find the number of units needed to be sold to make a net profit of $\$45\,000$. Round to the nearest whole unit.

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